4-2-1: Examining IPv4 Address Classes

At the end of this episode, I will be able to:

1. Identify and explain the structure of IPv4 address classes and importance, given a scenario.

Learner Objective: *Identify and explain the structure of IPv4 address classes and importance, given a scenario*

Description: In this episode, the learner will examine the IPv4 addressing class structure. We will explore classes ranges, network and hosts capacities, common usage and more.

- What are the address ranges of each IPv4 class?
    - Class A - 1.0.0.0 to 126.255.255.255 (Default Subnet Mask = 255.0.0.0 or /8)
    - Class B - 128.0.0.0 to 191.255.255.255 (Default Subnet Mask = 255.255.0.0 or /16)
    - Class C - 192.0.0.0 to 223.255.255.255 (Default Subnet Mask = 255.255.255.0 or /24)
    - Class D - 224.0.0.0 to 239.255.255.255 (not applicable, no assign subnet mask)
    - Class E - 240.0.0.0 to 255.255.255.255 (not applicable, no assign subnet mask)
- How many networks and hosts can each IPv4 class support?
    - Class A - Supports 128 networks with approximately 16 million hosts each
    - Class B - Supports 16,384 networks with 65,534 hosts each
    - Class C - Supports over 2 million networks with up to 254 hosts each
    - Class D - Not used for standard host addresses (multicast)
    - Class E - Reserved for experimental; not for general use
- What are the common uses for each IPv4 class?
    - Class A - Large organizations or networks requiring a vast number of IP addresses
    - Class B - Medium-sized networks such as universities or regional services
    - Class C - Small businesses and local area networks (LANs)
    - Class D - Multicast groups
    - Class E - Experimental purposes
- Can you provide examples of where each IPv4 class is typically used?
    - Class A - Major global enterprises, internet service providers
    - Class B - Educational institutions, larger corporations
    - Class C - Small companies, office networks
    - Class D - Streaming services, multi-user games
    - Class E - Network protocol testing, future use experimentation
- Are there IPv4 addresses that have been reserved or special use?
    - 0.0.0.0/8 - reserved by IANA
        - See IETF's *Special-Purpose Address Registries* RFC 6890 for more detailed information: https://datatracker.ietf.org/doc/html/rfc6890
    - 127.0.0.0/8
        - See IETF's *Special-Purpose Address Registries* RFC 6890 for more detailed information: https://datatracker.ietf.org/doc/html/rfc6890
    - 169.254.0.0/16
        - See IETF's *Special USe IPv4 Addresses* RFC 5735 for more detailed information: https://datatracker.ietf.org/doc/html/rfc5735

- Additional Resources:
    - If applicable